Grid Puzzle I

Time limit: 1.00 s **Memory limit:** 512 MB

There is an $n \times n$ grid, and your task is to choose from each row and column some number of squares. How can you do that?

Input

The first input line has an integer n: the size of the grid. The rows and columns are numbered $1, 2, \ldots, n$.

The next line has n integers $a_1, a_2, \ldots, a_n$: You must choose exactly a_i squares from the ith row.

The las line has n integers $b_1, b_2, \ldots, b_n$: You must choose exactly b_j squares from the jth column.

Output

Print n lines describing which squares you choose (`X` means that you choose a square, `.` means that you don't choose it). You can print any valid solution.

If it is not possible to satisfy the conditions print only -1.

Constraints

- $1 \le n \le 50$
- $0 \le a_i \le n$
- $0 \le b_j \le n$

Example

Input

```
5
0 1 3 2 0
1 2 2 0 1
```

Output

```
.....
..X..
.XX.X
XX...
.....
```